

Research note: practising LLD with feedback you can check

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1. The learner problem

I learned low-level design the way most students around me did: a YouTube walkthrough of Parking Lot, then the solution in a GitHub repository. When my design looked different from theirs, I could not tell whether it was wrong or just different. For DSA, LeetCode's judge settles that in a second. **For LLD there is no judge**, so I either copied the reference or guessed.

That gap costs a learner in three ways.

1. With one reference to compare against, a valid alternative looks like a mistake and a copied design looks like a good one.
2. The follow-up never gets practised, although real rounds test how a design absorbs change. [workat.tech](#) lists "Code should easily accommodate new requirements with minimal changes" as an expectation in machine coding rounds at Flipkart, Uber and Swiggy [1], and Hello Interview notes that "most interviews include a small follow-up requirement" [2]. Solo practice almost never includes one.
3. Nothing remembers my mistakes. Every attempt starts from zero, so the same weakness (one class doing everything) comes back problem after problem.

2. What exists today

Tool	What the learner submits	Feedback	Gap for a learner like me
LeetCode "Design Parking System" (Easy) [3]	Code for <code>addCar(carType)</code>	Pass or fail on tests	Checks behaviour, not design. The whole problem fits in three counters.
awesome-low-level-design, GitHub, 27k stars [4]	Nothing	Reference solutions in several languages	Compare-to-reference only. The most popular LLD resource has no feedback at all.
AlgoMaster LLD, 33 problems [5]	Nothing	Written solutions; premium tier	Same: learn by reading, check by comparing.
Codemia OOD, 50+ problems [6]	A design on the platform	Editorial solutions; its AI feedback is advertised for system design	Unclear whether OOD designs get AI feedback.
Hello Interview LLD Guided Practice [7]	Answers to each interview step	"Personalized feedback"	Premium and desktop only.
A general chatbot (ChatGPT, Claude)	Anything	A score and free text if asked	See the experiment below.
A human mock interview	A live discussion	The best feedback there is	Expensive and scarce.

Learners describe the same problem. A Hello Interview community thread is titled "Preparing for Low-Level Object-Oriented Design (LLD) feels incredibly non-trivial and disjointed" [8], and [r/leetcode](#) has threads such as "Master LLD for interviews and looking for study buddies" [9].

3. A small experiment: what does an LLM actually tell you?

I gave the same flawed Parking Lot design (one class that parks, prices, takes payment and prints receipts) to `deepseek/deepseek-v4.1-flash` five times in each of three ways, and then repeated the whole run with the model's reasoning switched off (`npm run experiment:consistency`):

Setup	Run 1 (reasoning on)	Run 2 (reasoning off)
A. "Rate it from 1 to 10", default temperature, like a chat app	5, 6, 5, 5, 4	6, 6, 6, 6, 6
B. The same question at temperature 0	5, 5, 5, 5, 5	6, 6, 6, 6, 6
C. DesignKata's rubric at temperature 0	5 of 6 criteria gave the same level every time; 46 of 47 quotes found word for word	5 of 6 the same every time; 66 of 68 quotes found

This surprised me: **a bare score is easy to make repeatable**. Temperature 0 did it, and so did turning reasoning off. But the score moved from 5 to 6 when one setting changed, and "6 out of 10" still does not say *what* to change. The rubric named the same two faults (one class doing everything, pricing that cannot vary) at level 1 in 19 of 20 judgments, and code could verify 112 of its 115 quotes. What a learner lacks is **feedback that says what to change and can be checked against their own words**. That finding shaped the product more than any competitor did. Raw output is in [docs/experiments/](#).

4. Gaps worth solving

1. There is no judge. Learners need a verdict for each quality of a design, rather than one number or one reference answer.
2. Feedback cannot be checked. An LLM's claims about a design should point at the learner's own words, and something should check those pointers.
3. Change is neither practised nor measured, although it is what interviewers probe.
4. Nothing tracks a learner's weaknesses across attempts.
5. The tools that do give feedback sit behind paid tiers, while CipherSchools' learners are students in India.

5. Product direction: DesignKata

The loop: pick a problem, design, submit, get feedback, then revise or take the twist.

- **Submission:** five short guided notes (requirements, entities, flows, trade-offs, edge cases) plus a class diagram written as Mermaid text. This is the smallest format that shows both the reasoning and the structure. It is also machine-readable, which a drawing is not, and it takes less time than full code.
- **Feedback you can check:** deterministic rules catch structural problems instantly and for free. An LLM then judges six rubric criteria on anchored levels from 1 to 4. It must quote the learner for every judgment, and code verifies that each quote really exists. There is no overall score.
- **The twist:** after the first review, the learner gets an interviewer-style new requirement. DesignKata diffs the two class diagrams and reports the blast radius: how many existing classes had to change. This is how a design can be judged by how it absorbs change, instead of by how close it is to a reference.
- **History:** every version is kept, with per-criterion levels over time and the weaknesses that keep coming back.
- **Not in the MVP:** accounts, code or hand-drawn submissions, an admin panel, leaderboards.

Sources:

- [1] [workat.tech](#), "What is a Machine Coding Round?"
- [2] Hello Interview, "Low Level Design in a Hurry"
- [3] [leetcode.com/problems/design-parking-system](#)
- [4] [github.com/ashishps1/awesome-low-level-design](#) (27,064 stars on 26 Sep 2026)
- [5] [algomaster.io/learn/lld](#)
- [6] [codemia.io/object-oriented-design](#)
- [7] [hellointerview.com/practice/low-level-design](#)
- [8] [hellointerview.com/community/discussion/low-level-design-preparation](#)
- [9] [r/leetcode](#), "Master LLD for interviews and looking for study buddies"